

UCS701: THEORY OF COMPUTATION

L	T	P	Cr
3	1	0	3.5

Course Objectives: This course introduces basic theory of computer science and formal methods of computation. The course exposes students to the computability theory, as well as to the complexity theory.

Regular Languages: Alphabets, Language, Regular Expression, Definitions of Finite State Machine, Transition Graphs, Deterministic & Non-deterministic Finite State Machines, Regular Grammar, Thompson's Construction to Convert Regular Expression to NDFA & Subset Algorithm to convert NDFA to DFA, Various recent development in the Conversion of Regular Expression to NFA, Minimization of DFA, Finite State Machine with output- Moore machine and Melay Machine, Conversion of Moore machine to Melay Machine & Vice-Versa.

Properties of Regular languages: Conversion of DFA to Regular Expression, Pumping Lemma, Properties and Limitations of Finite state machine, Decision properties of Regular Languages, Application of Finite Automata.

Context Free Grammar and Push Down Automata: Context Free Grammar, Derivation tree and Ambiguity, Application of Context free Grammars, Chomsky and Greibach Normal form, Properties of context free grammar, CKY Algorithm, Decidable properties of Context free Grammar, Pumping Lemma for Context free grammar, Push down Stack Machine, Design of Deterministic and Non-deterministic Push-down stack.

Turing Machine: Turing machine definition and design of Turing Machine, Variations of Turing Machines, combining Turing machine, Universal Turing Machine, Post Machine, Chomsky Hierarchy, Post correspondence problem, Halting problem, Turing decidability.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Comprehend regular languages and finite automata and develop ability to provide the equivalence between regular expressions, NFAs, and DFAs.
2. Disambiguate context-free grammars and understand the concepts of context-free languages and pushdown automata.
3. Analyse and design efficient Turing Machines.
4. Distinguish different computing languages and classify their respective types.

Text Books:

1. Hopcroft E. J., Ullman D. J. and Motwani R., Introduction to Automata Theory, Languages and Computation, Pearson Education (2007) 3rd ed.
2. Martin C. J., Introduction to Languages and the Theory of Computation, McGraw-Hill Higher Education (2011) 4th ed.
3. Lewis R. H., Papadimitriou H. C., Elements of the Theory of Computation, Prentice Hall (1998) 2nd ed.

Reference Books:

1. Cohen A. I. D., Introduction to Computer Theory, Wiley (1997) 2nd ed.
2. Sipser M., Introduction to the Theory of Computation, Cengage Learning (2013) 3rd ed.